Temasek POLYTECHNIC

SCHOOL OF APPLIED SCIENCE

PROJECT REPORT

Project Title	Development of an Innovative LiGrA®-Based Substrate Technology to Enhance Agricultural Productivity
Project Ref	**TPIH/PCP/25/98**

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Project Closing Report

TP/PCP/25/98

Title: Development of an Innovative LiGrA®-Based Substrate Technology to Enhance Agricultural Productivity

1. Background

Modern horticulture and landscaping systems remain heavily dependent on peat moss and perlite as primary growth substrates. While effective agronomically, these materials present significant environmental and supply-chain challenges that are increasingly incompatible with current climate, sustainability, and circular-economy mandates.

Peat extraction leads to the irreversible destruction of carbon-rich wetlands, loss of biodiversity, and the release of long-sequestered greenhouse gases, contributing substantially to climate change. At the same time, perlite production relies on energy-intensive thermal expansion processes (approximately 900–1000 °C), resulting in high embodied energy and associated carbon emissions. The global transport of perlite further compounds its environmental footprint and exposes growers to supply volatility.

As agriculture intensifies, the demand for growth substrates is rising. However, growers face mounting pressure to adopt low-carbon, circular, and regenerative practices—creating an urgent need for alternatives that meet both performance standards and climate targets.

LiGrA® is an advanced material engineered from recycled inorganic waste streams, embodying the principles of the circular economy. mGRO, a type of LiGrA® used for agriculture, mirrors perlite's most valued traits; lightweight & porous, inert and stable, locally producible and support growth of micro-organisms.

The purpose of the project is to develop and optimize mGRO-based substrate formulations for landscaping and outdoor farming applications.

- Incorporating mGRO into soil-based substrates as a sustainable replacement for conventional aggregates
- Integrate Effective Microorganisms (EM) into substrate to enhance biological functionality and plant performance
- Evaluate plant growth performance and substrate functionality across multiple mGRO ratios
- Identify an optimized formulation that balances productivity, sustainability and scalability

2. Experimental Design

    2.1 Effective Microorganism (EM) and mGRO

    - Control A (Soil)
    - Control B (Soil + Microbial Treatment)

- Treatment 1 (Soil + mGRO soaked with microbes + Microbial Treatment)
- Treatment 2 (Soil + Raw mGRO + Microbial Treatment)

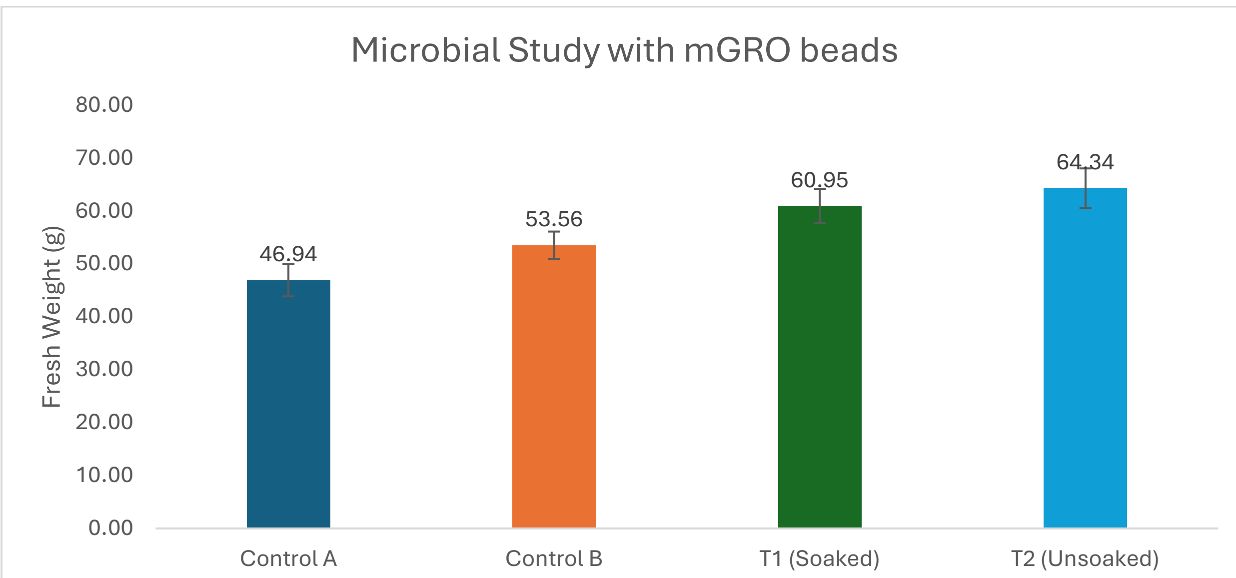
mGRO beads were first soaked with EM1 (1 part to 500 parts water) for 48 hours before use as substrate.

2.2 mGRO Substrate Study

- Control A (Peat Moss + Perlite)
- Composition 1 (60% Coco Peat + 30% mGRO beads + 10% Vermicompost)
- Composition 2 (45% Coco Peat + 45% mGRO beads + 10% Vermicompost)

3. Results and Discussion

3.1 EM and mGRO



Graph 1: Comparison chart for microbial study with mGRO beads (n=32 for each treatment)

Comparing yield of Control A (46.94±3.06g) to Control B (53.56±2.58g), it can be observed that microbial activity does indeed aid with plant growth, with 14.1% higher fresh weight at harvest. T1 (60.95±3.25g) and T2 (64.34±3.73g) also outperformed control A, with 29.8% and 37.1% increase in yield of Pak Choi compared to the control which show the potential of microbial activity in boosting plant yields.

Incorporation of mGRO beads into the substrate promising results and has shown significant improvement in plant growth. With both mGRO treatments having higher fresh weights at harvest compared to Control B, with T1 and T2 having 13.8% and 20.1% higher yields respectively. This establishes the ability of mGRO beads in aiding microbial growth and activity in the substrate, which then led to the enhanced yield obtained from these two treatments. A possible reason is that surface of mGRO beads allow for microbes to attach and grow and hence promoting the microbial ecosystem within the substrate.

However, the presoaked mGRO beads in T1 yielded a lower fresh weight compared to T2, which could indicate that presoaking the mGRO beads has caused the microbes to oversaturate. Possible ways to circumvent this is to lower the dosage of EM1 during presoaking phase or during the treatment.

3.2 mGRO Substrate Study

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JOB REF. NO.: J2026-03-06-014
Page 2 of 2

TEST RESULTS:

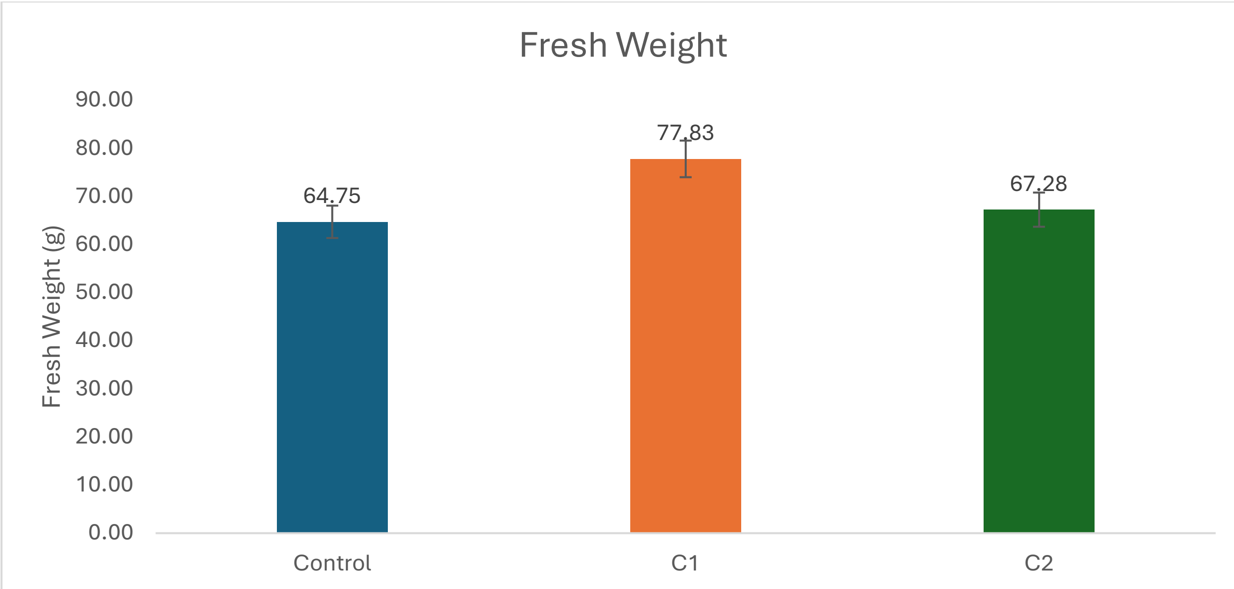
DETERMINATION OF pH VALUE (BS 1377-3:2018 + A1:2021 Sect 12)

SAMPLE DESCRIPTION	pH			
	30 min	1 Day	3 Days	7 Days
100 % COCOPEAT	6.09 @ 23.8 °C	6.08 @ 23.5 °C	6.06 @ 22.6 °C	6.07 @ 23.7 °C
100 % LIGRA	10.04 @ 24.1 °C	9.86 @ 23.7 °C	9.83 @ 22.6 °C	9.59 @ 23.8 °C
50 % COCOPEAT + 50 % LIGRA	6.87 @ 24.4 °C	6.96 @ 23.5 °C	7.00 @ 22.3 °C	7.04 @ 23.7 °C

Picture 1: pH tests results of Ligra (mGRO) from accredited lab

Peat moss is an unsustainable source of substrate due to its harvest destroying peatlands, which are some of the world's most effective carbon sinks. Peatlands are also an important part of habitat and ecosystem and take thousands of years to regenerate, hence finding a suitable and sustainable replacement for peat moss is crucial.

Looking at the mGRO beads pH, it seems unsuitable for use in agriculture at first glance, due to the high pH of the mGRO beads. However, using it in tandem with coco peat helps to stabilise the pH of the substrate and enable it to be used as a substrate.



Graph 2: Comparison chart for substrate study with mGRO beads (n=32 for each treatment)

For the substrate formulation study, different compositions of mGRO beads with coco peats were explored. Both substrate formulas in C1 (77.83±3.81g) and C2 (67.28±3.55g) yielded a higher average fresh weight compared to the control (64.75±3.37g), indicating that mGRO composition with coco peat can work well as a substrate replacement to peat moss. However, higher percentage of mGRO beads might affect the stability of pH in the substrate, which can affect the growth of plants so an optimal composition has to be explored.

Currently, C1, with 60% of cocopeat to 30% of mGRO beads with 10% vermicompost to provide nutrients is sufficient in improving the yield of plants by more than 15% whilst being the more sustainable option.

4. Conclusion and Recommendations

The microbial study had shows promising results in improving plant growth in microbial conditions. This may need to be verified with a scanning electron microscope (SEM), assuming that the microbes are attached to the surface of the mGRO beads.

Substrate formulation with mGRO also indicated positive results, performing better than peat moss; a secondary test may need to be carried out to verify the effectiveness of growth without vermicompost and to better optimise the formula for a sustainable substrate with mGRO beads.

From the results of the two experiments, it can be observed that mGRO beads are useful in agriculture; as a medium for microbial attachment and as a part of substrate. mGRO beads can be further explored to expand its application in agriculture.

5. Appendix

Photo 2: Picture comparing the plants at harvest for microbial study, from left to right (T2, Control A, Control B, T1)

Photo 3: Picture comparing the plants at harvest for substrate study, from left to right (control, T1, T2)